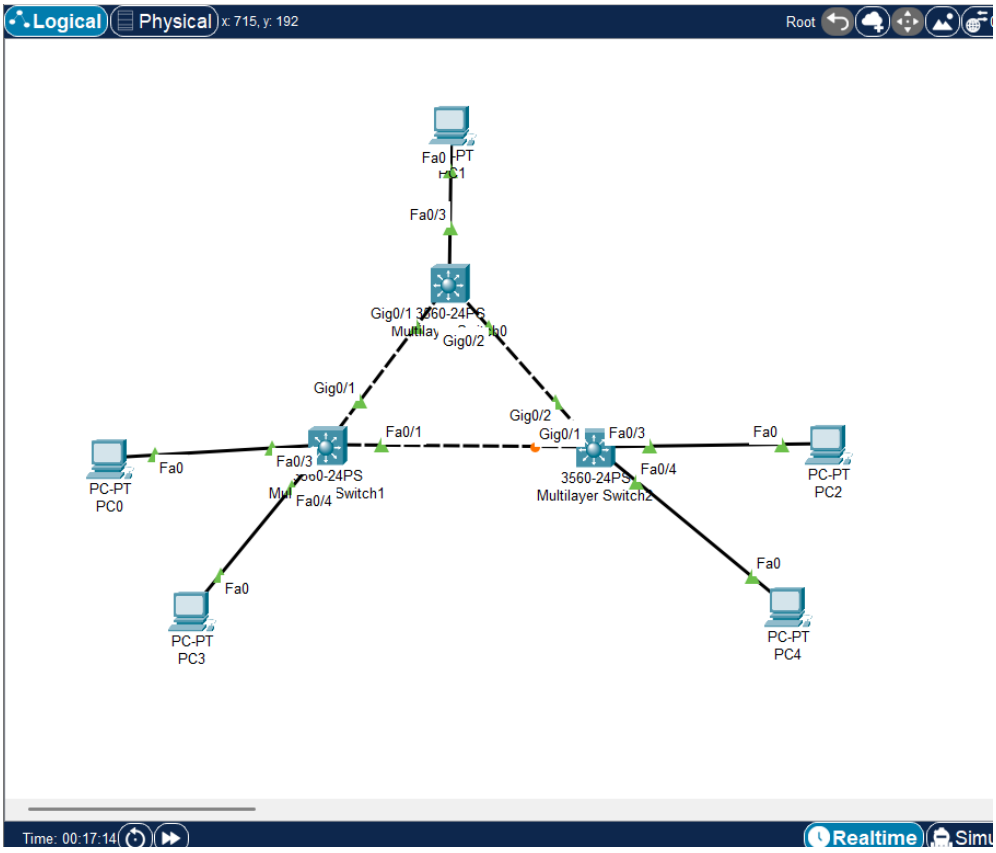
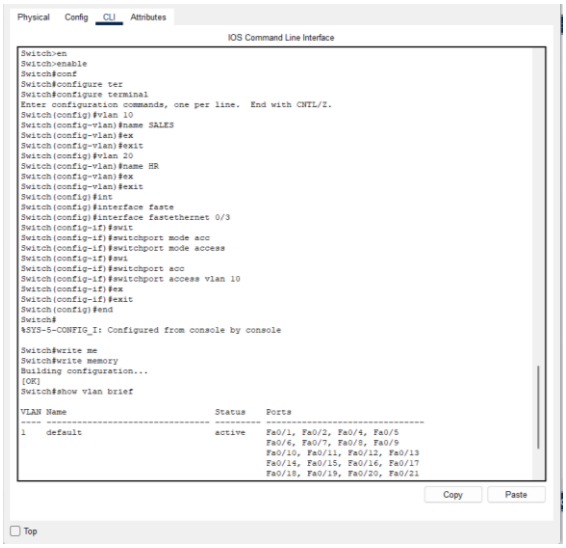
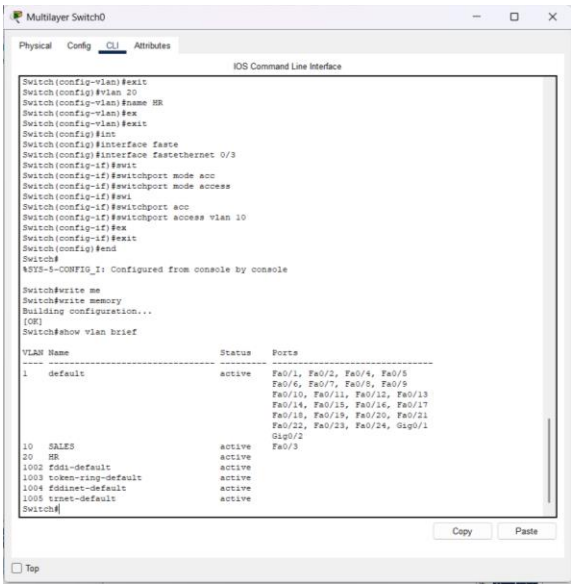
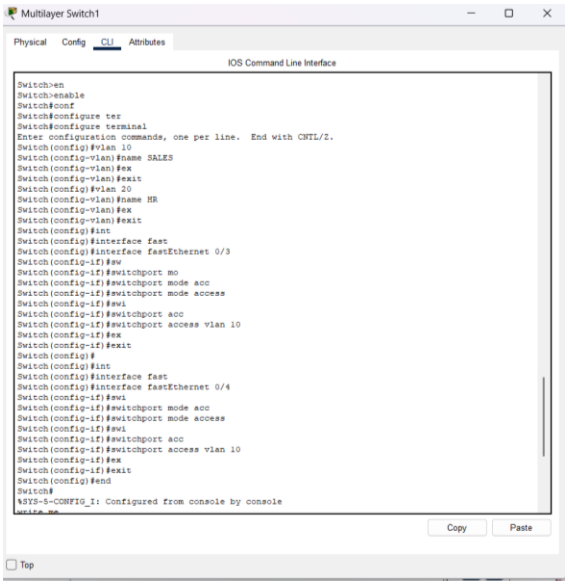
Task 1: Build the topology

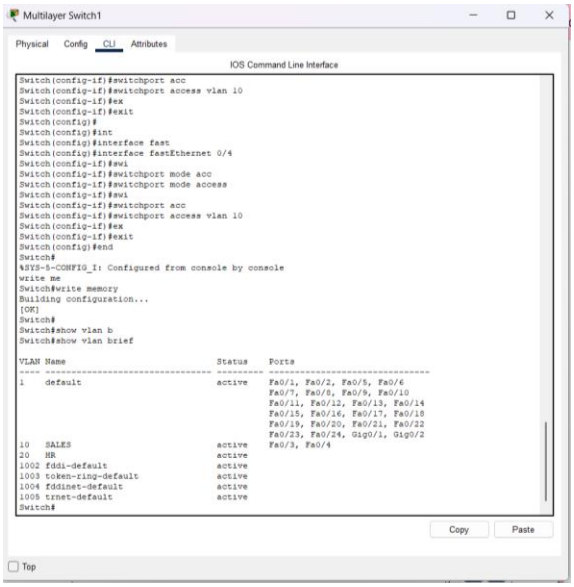


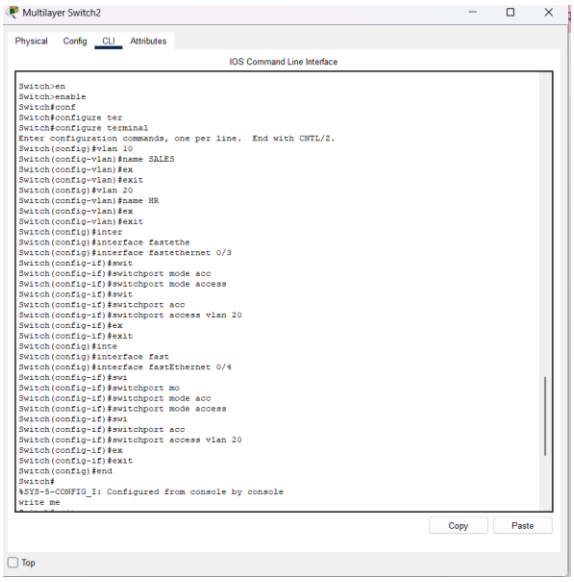
Task 2: Configure VLANs

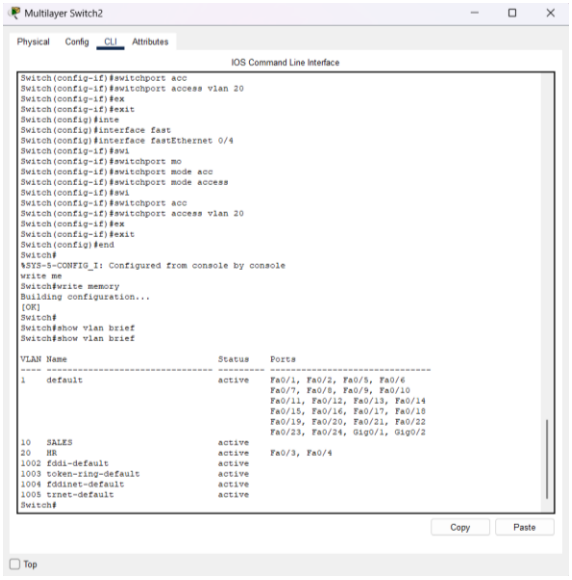

Physical
Config
CLI
Attributes
IOS Command Line Interface
Switch>en
Switch>enable
Switch#conf
Switch#configure ter
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name SALES
Switch(config-vlan)#ex
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name HR
Switch(config-vlan)#ex
Switch(config-vlan)#exit
Switch(config)#int
Switch(config)#interface faste
Switch(config)#interface fastethernet 0/3
Switch(config-if)#swit
Switch(config-if)#switchport mode acc
Switch(config-if)#switchport mode access
Switch(config-if)#swi
Switch(config-if)#switchport acc
Switch(config-if)#switchport access vlan 10
Switch(config-if)#ex
Switch(config-if)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
Switch#write me
Switch#write memory
Building configuration...
[OK]
Switch#show vlan brief
VLAN Name Status Ports
1 default active Fa0/1, Fa0/2, Fa0/4, Fa0/5
Fa0/6, Fa0/7, Fa0/8, Fa0/9
Fa0/10, Fa0/11, Fa0/12, Fa0/13
Fa0/14, Fa0/15, Fa0/16, Fa0/17
Fa0/18, Fa0/19, Fa0/20, Fa0/21
Copy
Paste
Top


Multilayer Switch0
Physical
Config
CLI
Attributes
IOS Command Line Interface
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name HR
Switch(config-vlan)#ex
Switch(config-vlan)#exit
Switch(config)#int
Switch(config)#interface faste
Switch(config)#interface fastethernet 0/3
Switch(config-if)#swit
Switch(config-if)#switchport mode acc
Switch(config-if)#switchport mode access
Switch(config-if)#swi
Switch(config-if)#switchport acc
Switch(config-if)#switchport access vlan 10
Switch(config-if)#ex
Switch(config-if)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
Switch#write me
Switch#write memory
Building configuration...
[OK]
Switch#show vlan brief
VLAN Name Status Ports
1 default active Fa0/1, Fa0/2, Fa0/4, Fa0/5
Fa0/6, Fa0/7, Fa0/8, Fa0/9
Fa0/10, Fa0/11, Fa0/12, Fa0/13
Fa0/14, Fa0/15, Fa0/16, Fa0/17
Fa0/18, Fa0/19, Fa0/20, Fa0/21
Fa0/22, Fa0/23, Fa0/24, Gig0/1
Gig0/2
10 SALES active Fa0/3
20 HR active
1002 fddi-default active
1003 token-ring-default active
1004 fddinet-default active
1005 trnet-default active
Switch#
Copy
Paste
Top

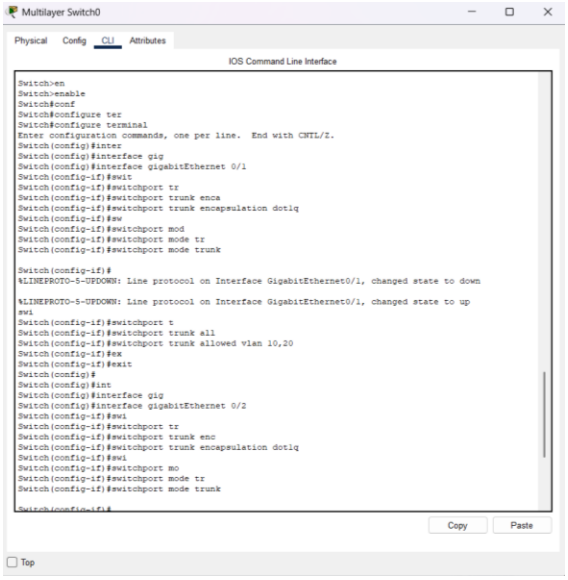

Multilayer Switch1
Physical
Config
CLI
Attributes
IOS Command Line Interface
Switch>en
Switch>enable
Switch#conf
Switch#configure ter
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name SALES
Switch(config-vlan)#ex
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name HR
Switch(config-vlan)#ex
Switch(config-vlan)#exit
Switch(config)#int
Switch(config)#interface fast
Switch(config)#interface fastEthernet 0/3
Switch(config-if)#sw
Switch(config-if)#switchport mo
Switch(config-if)#switchport mode acc
Switch(config-if)#switchport mode access
Switch(config-if)#wi
Switch(config-if)#switchport acc
Switch(config-if)#switchport access vlan 10
Switch(config-if)#ex
Switch(config-if)#exit
Switch(config)#
Switch(config)#int
Switch(config)#interface fast
Switch(config)#interface fastEthernet 0/4
Switch(config-if)#swi
Switch(config-if)#switchport mode acc
Switch(config-if)#switchport mode access
Switch(config-if)#wi
Switch(config-if)#switchport acc
Switch(config-if)#switchport access vlan 10
Switch(config-if)#ex
Switch(config-if)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
Copy
Paste
Top

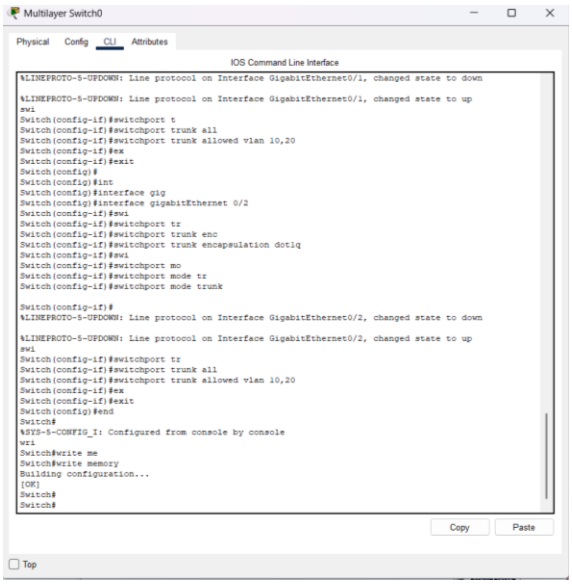

Multilayer Switch1
Physical
Config
CLI
Attributes
IOS Command Line Interface
Switch(config-if)#switchport acc
Switch(config-if)#switchport access vlan 10
Switch(config-if)#ex
Switch(config-if)#exit
Switch(config)#
Switch(config)#int
Switch(config)#interface fast
Switch(config)#interface fastEthernet 0/4
Switch(config-if)#swi
Switch(config-if)#switchport mode acc
Switch(config-if)#switchport mode access
Switch(config-if)#wi
Switch(config-if)#switchport acc
Switch(config-if)#switchport access vlan 10
Switch(config-if)#ex
Switch(config-if)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
write me
Switch#write memory
Building configuration...
[OK]
Switch#
Switch#show vlan b
Switch#show vlan brief
VLAN Name Status Ports
1 default active Fa0/1, Fa0/2, Fa0/5, Fa0/6
Fa0/7, Fa0/8, Fa0/9, Fa0/10
Fa0/11, Fa0/12, Fa0/13, Fa0/14
Fa0/15, Fa0/16, Fa0/17, Fa0/18
Fa0/19, Fa0/20, Fa0/21, Fa0/22
Fa0/23, Fa0/24, Gig0/1, Gig0/2
10 SALES active Fa0/3, Fa0/4
20 HR active
1002 fddi-default active
1003 token-ring-default active
1004 fddinet-default active
1005 trnet-default active
Switch#
Copy
Paste
Top

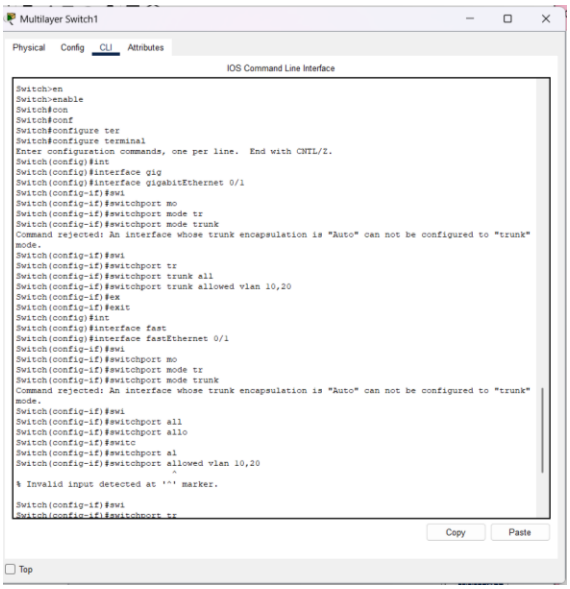


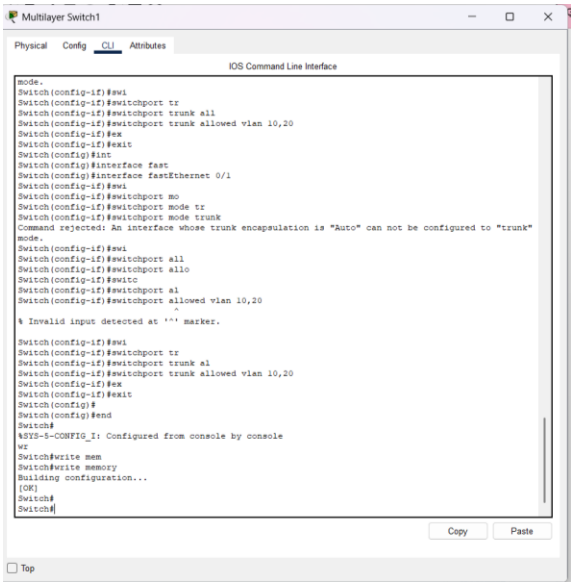


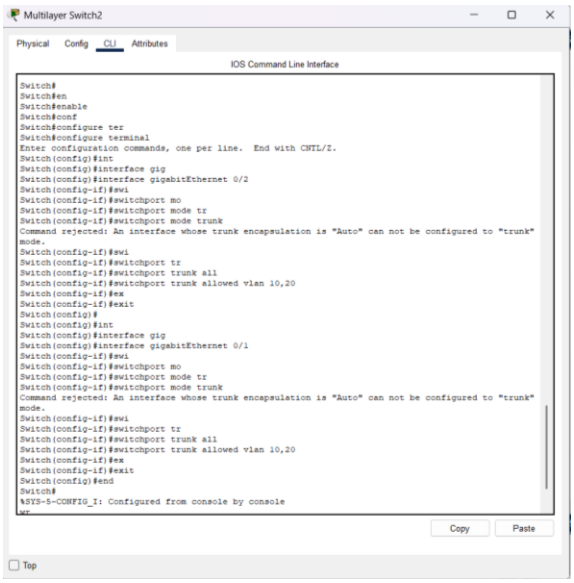
Task 3: Configure trunk links

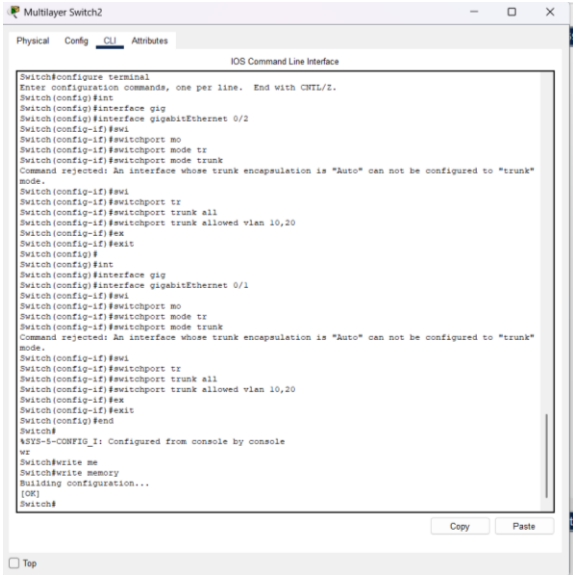

Multilayer Switch0
Physical
Config
CLI
Attributes
IOS Command Line Interface
Switch>en
Switch>enable
Switch#conf
Switch#configure ter
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#inter
Switch(config)#interface gig
Switch(config)#interface gigabitEthernet 0/1
Switch(config-if)#swit
Switch(config-if)#switchport tr
Switch(config-if)#switchport trunk enca
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#sw
Switch(config-if)#switchport mod
Switch(config-if)#switchport mode tr
Switch(config-if)#switchport mode trunk
Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
swi
Switch(config-if)#switchport t
Switch(config-if)#switchport trunk all
Switch(config-if)#switchport trunk allowed vlan 10,20
Switch(config-if)#ex
Switch(config-if)#exit
Switch(config)#
Switch(config)#int
Switch(config)#interface gig
Switch(config)#interface gigabitEthernet 0/2
Switch(config-if)#swi
Switch(config-if)#switchport tr
Switch(config-if)#switchport trunk enc
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#swi
Switch(config-if)#switchport mo
Switch(config-if)#switchport mode tr
Switch(config-if)#switchport mode trunk
Copy
Paste
Top

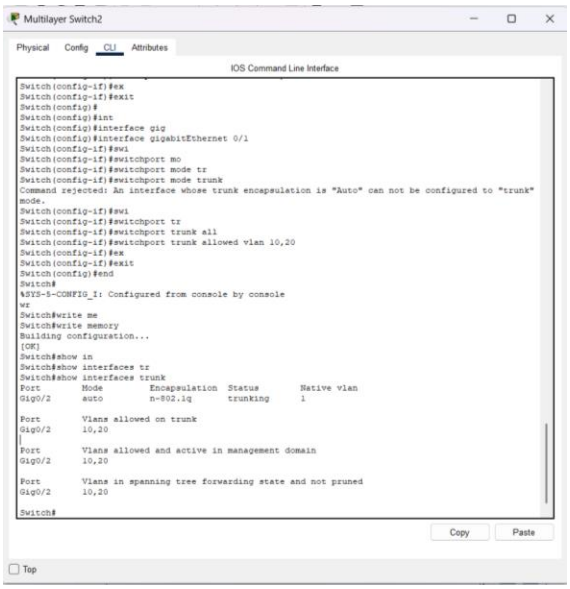

Multilayer Switch0
Physical
Config
CLI
Attributes
IOS Command Line Interface
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
swi
Switch(config-if)#switchport t
Switch(config-if)#switchport trunk all
Switch(config-if)#switchport trunk allowed vlan 10,20
Switch(config-if)#ex
Switch(config-if)#exit
Switch(config)#
Switch(config)#int
Switch(config)#interface gig
Switch(config)#interface gigabitEthernet 0/2
Switch(config-if)#swi
Switch(config-if)#switchport tr
Switch(config-if)#switchport trunk enc
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#swi
Switch(config-if)#switchport mo
Switch(config-if)#switchport mode tr
Switch(config-if)#switchport mode trunk
Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
swi
Switch(config-if)#switchport tr
Switch(config-if)#switchport trunk all
Switch(config-if)#switchport trunk allowed vlan 10,20
Switch(config-if)#ex
Switch(config-if)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
wri
Switch#write me
Switch#write memory
Building configuration...
[OK]
Switch#
Switch#
Copy
Paste
Top


Multilayer Switch1
Physical
Config
CLI
Attributes
IOS Command Line Interface
Switch>en
Switch>enable
Switch#con
Switch#conf
Switch#configure ter
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int
Switch(config)#interface gig
Switch(config)#interface gigabitEthernet 0/1
Switch(config-if)#swi
Switch(config-if)#switchport mo
Switch(config-if)#switchport mode tr
Switch(config-if)#switchport mode trunk
Command rejected: An interface whose trunk encapsulation is "Auto" can not be configured to "trunk"
mode.
Switch(config-if)#swi
Switch(config-if)#switchport tr
Switch(config-if)#switchport trunk all
Switch(config-if)#switchport trunk allowed vlan 10,20
Switch(config-if)#ex
Switch(config-if)#exit
Switch(config)#int
Switch(config)#interface fast
Switch(config)#interface fastEthernet 0/1
Switch(config-if)#swi
Switch(config-if)#switchport mo
Switch(config-if)#switchport mode tr
Switch(config-if)#switchport mode trunk
Command rejected: An interface whose trunk encapsulation is "Auto" can not be configured to "trunk"
mode.
Switch(config-if)#swi
Switch(config-if)#switchport all
Switch(config-if)#switchport allo
Switch(config-if)#switc
Switch(config-if)#switchport al
Switch(config-if)#switchport allowed vlan 10,20
^
% Invalid input detected at '^' marker.
Switch(config-if)#swi
Switch(config-if)#switchport tr
Copy
Paste
Top


Multilayer Switch1
Physical
Config
CLI
Attributes
IOS Command Line Interface
mode.
Switch(config-if)#swi
Switch(config-if)#switchport tr
Switch(config-if)#switchport trunk all
Switch(config-if)#switchport trunk allowed vlan 10,20
Switch(config-if)#ex
Switch(config-if)#exit
Switch(config)#int
Switch(config)#interface fast
Switch(config)#interface fastEthernet 0/1
Switch(config-if)#swi
Switch(config-if)#switchport mo
Switch(config-if)#switchport mode tr
Switch(config-if)#switchport mode trunk
Command rejected: An interface whose trunk encapsulation is "Auto" can not be configured to "trunk"
mode.
Switch(config-if)#swi
Switch(config-if)#switchport all
Switch(config-if)#switchport allo
Switch(config-if)#switc
Switch(config-if)#switchport al
Switch(config-if)#switchport allowed vlan 10,20
^
% Invalid input detected at '^' marker.
Switch(config-if)#swi
Switch(config-if)#switchport tr
Switch(config-if)#switchport trunk al
Switch(config-if)#switchport trunk allowed vlan 10,20
Switch(config-if)#ex
Switch(config-if)#exit
Switch(config)#
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
wr
Switch#write mem
Switch#write memory
Building configuration...
[OK]
Switch#
Switch#
Copy
Paste
Top

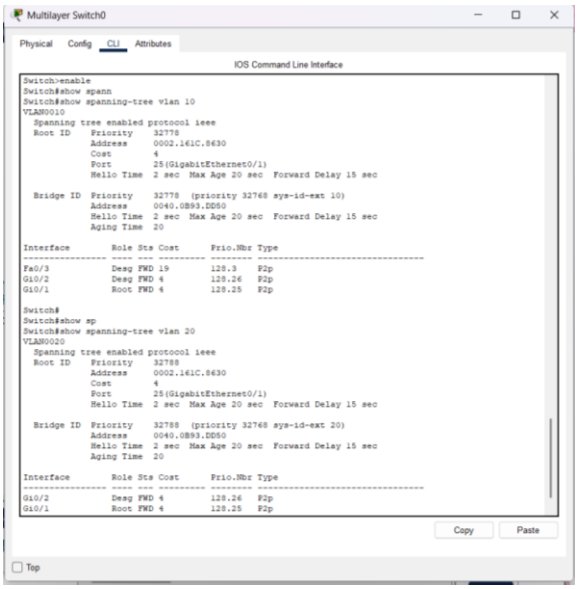

Multilayer Switch2
Physical
Config
CLI
Attributes
IOS Command Line Interface
Switch#
Switch#en
Switch#enable
Switch#conf
Switch#configure ter
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int
Switch(config)#interface gig
Switch(config)#interface gigabitEthernet 0/2
Switch(config-if)#swi
Switch(config-if)#switchport mo
Switch(config-if)#switchport mode tr
Switch(config-if)#switchport mode trunk
Command rejected: An interface whose trunk encapsulation is "Auto" can not be configured to "trunk"
mode.
Switch(config-if)#swi
Switch(config-if)#switchport tr
Switch(config-if)#switchport trunk all
Switch(config-if)#switchport trunk allowed vlan 10,20
Switch(config-if)#ex
Switch(config-if)#exit
Switch(config)#
Switch(config)#int
Switch(config)#interface gig
Switch(config)#interface gigabitEthernet 0/1
Switch(config-if)#swi
Switch(config-if)#switchport mo
Switch(config-if)#switchport mode tr
Switch(config-if)#switchport mode trunk
Command rejected: An interface whose trunk encapsulation is "Auto" can not be configured to "trunk"
mode.
Switch(config-if)#swi
Switch(config-if)#switchport tr
Switch(config-if)#switchport trunk all
Switch(config-if)#switchport trunk allowed vlan 10,20
Switch(config-if)#ex
Switch(config-if)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
wr
Copy
Paste
Top

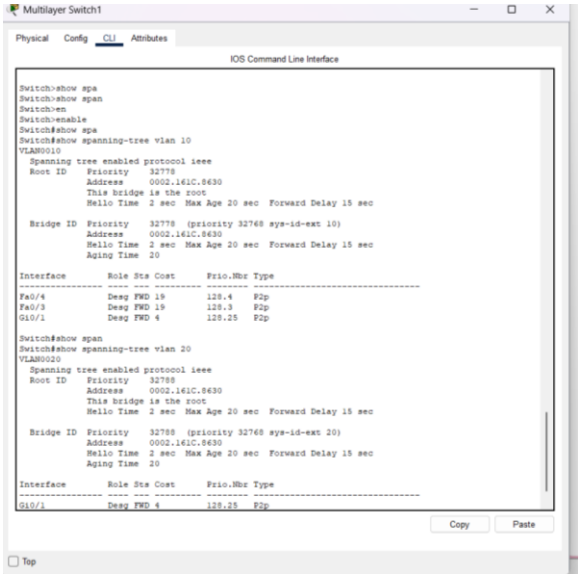

Multilayer Switch2
Physical
Config
CLI
Attributes
IOS Command Line Interface
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int
Switch(config)#interface gig
Switch(config)#interface gigabitEthernet 0/2
Switch(config-if)#swi
Switch(config-if)#switchport mo
Switch(config-if)#switchport mode tr
Switch(config-if)#switchport mode trunk
Command rejected: An interface whose trunk encapsulation is "Auto" can not be configured to "trunk"
mode.
Switch(config-if)#swi
Switch(config-if)#switchport tr
Switch(config-if)#switchport trunk all
Switch(config-if)#switchport trunk allowed vlan 10,20
Switch(config-if)#ex
Switch(config-if)#exit
Switch(config)#
Switch(config)#int
Switch(config)#interface gig
Switch(config)#interface gigabitEthernet 0/1
Switch(config-if)#swi
Switch(config-if)#switchport mo
Switch(config-if)#switchport mode tr
Switch(config-if)#switchport mode trunk
Command rejected: An interface whose trunk encapsulation is "Auto" can not be configured to "trunk"
mode.
Switch(config-if)#swi
Switch(config-if)#switchport tr
Switch(config-if)#switchport trunk all
Switch(config-if)#switchport trunk allowed vlan 10,20
Switch(config-if)#ex
Switch(config-if)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
wr
Switch#write me
Switch#write memory
Building configuration...
[OK]
Switch#
Copy
Paste
Top

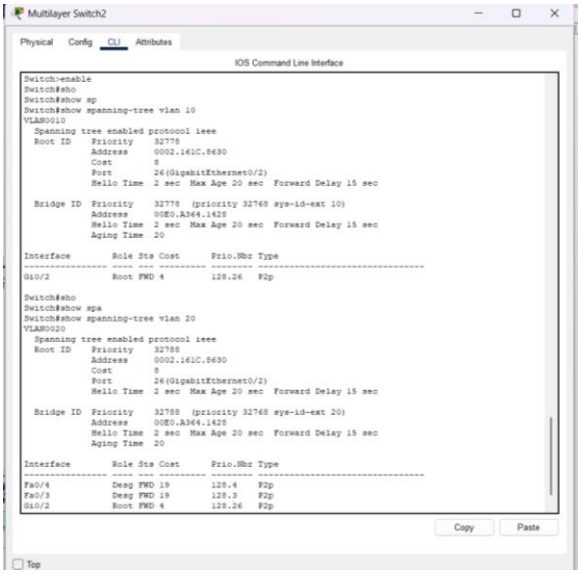


Task 4: Configure PC IP addresses

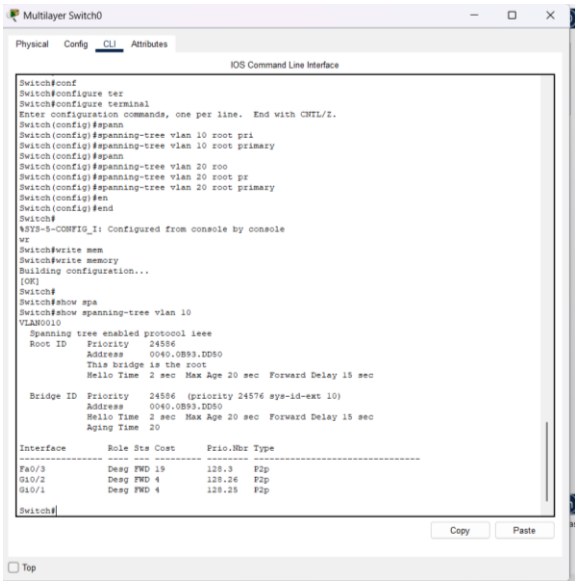
Task 5: Investigate default STP



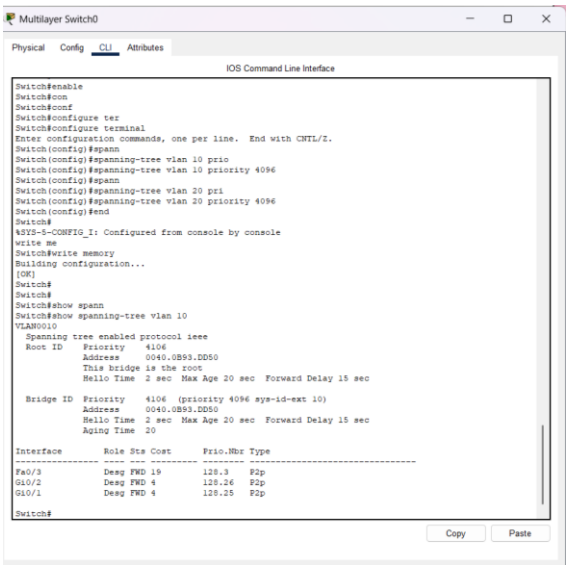


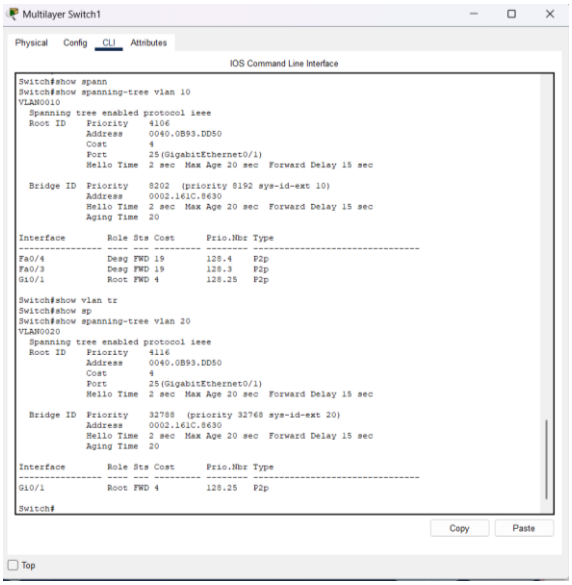


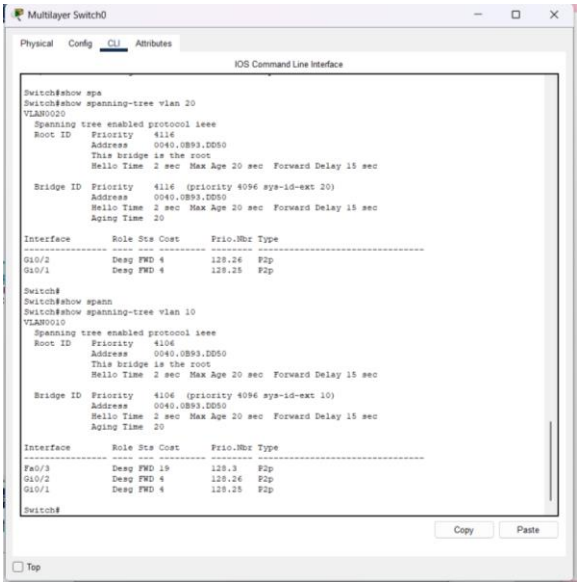
Task 6: Elect a Root Bridge



Task 7: Configure different priorities


Multilayer Switch0
Physical
Config
CLI
Attributes
IOS Command Line Interface
Switch#enable
Switch#con
Switch#conf
Switch#configure ter
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#spann
Switch(config)#spanning-tree vlan 10 prio
Switch(config)#spanning-tree vlan 10 priority 4096
Switch(config)#spann
Switch(config)#spanning-tree vlan 20 pri
Switch(config)#spanning-tree vlan 20 priority 4096
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
write me
Switch#write memory
Building configuration...
[OK]
Switch#
Switch#
Switch#show spann
Switch#show spanning-tree vlan 10
VLAN0010
  Spanning tree enabled protocol ieee
  Root ID   Priority   4106
              Address   0040.0B93.DD50
              This bridge is the root
              Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
  Bridge ID   Priority   4106 (priority 4096 sys-id-ext 10)
              Address   0040.0B93.DD50
              Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
              Aging Time 20
Interface   Role Sts Cost   Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/3   Desg FWD 19   128.3   P2p
Gi0/2   Desg FWD 4   128.26   P2p
Gi0/1   Desg FWD 4   128.25   P2p
Switch#
Copy
Paste


Multilayer Switch1
Physical
Config
CLI
Attributes
IOS Command Line Interface
Switch#show spann
Switch#show spanning-tree vlan 10
VLAN0010
  Spanning tree enabled protocol ieee
  Root ID   Priority   4106
              Address   0040.0B93.DD50
              Cost   4
              Port   25(GigabitEthernet0/1)
              Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
  Bridge ID   Priority   8202 (priority 8192 sys-id-ext 10)
              Address   0002.161C.8630
              Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
              Aging Time 20
Interface   Role Sts Cost   Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/4   Desg FWD 19   128.4   P2p
Fa0/3   Desg FWD 19   128.3   P2p
Gi0/1   Root FWD 4   128.25   P2p
Switch#show vlan tr
Switch#show sp
Switch#show spanning-tree vlan 20
VLAN0020
  Spanning tree enabled protocol ieee
  Root ID   Priority   4116
              Address   0040.0B93.DD50
              Cost   4
              Port   25(GigabitEthernet0/1)
              Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
  Bridge ID   Priority   32788 (priority 32768 sys-id-ext 20)
              Address   0002.161C.8630
              Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
              Aging Time 20
Interface   Role Sts Cost   Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Gi0/1   Root FWD 4   128.25   P2p
Switch#
Copy
Paste
Top



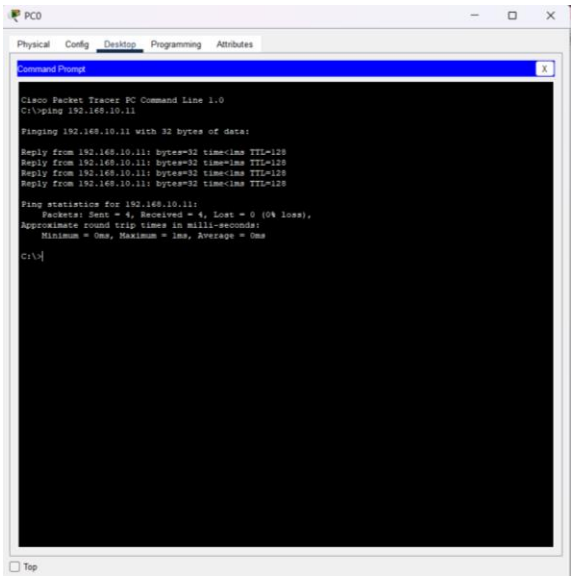
Task 8: Investigate redundant links

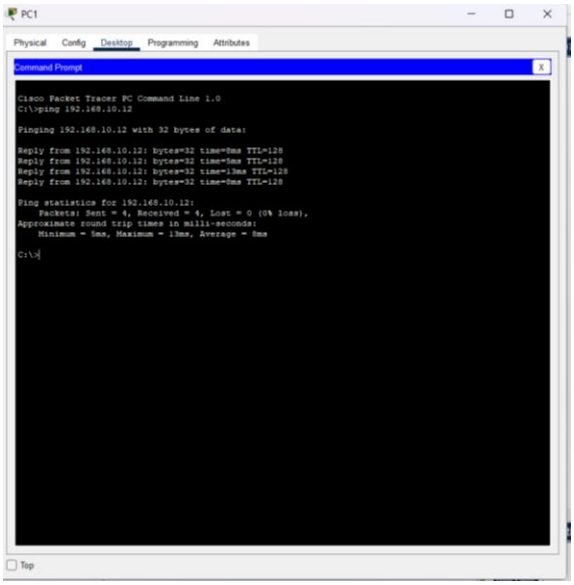
Identify the inter-switch link or port that STP blocks for each VLAN. Explain why a redundant path is necessary and why it must not forward traffic simultaneously with every other path.

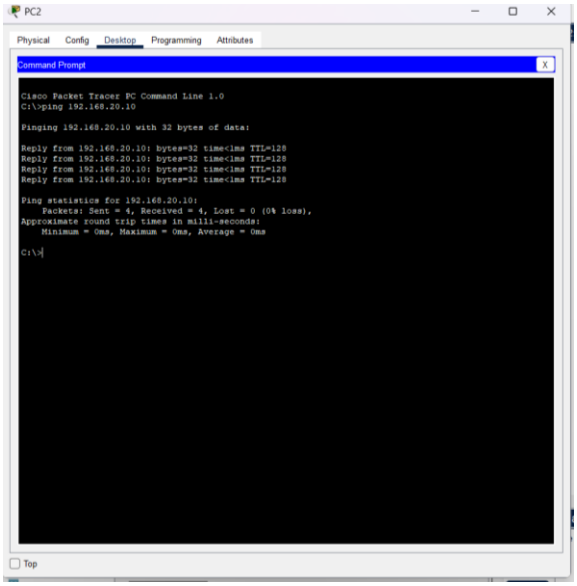
They provide automatic backup links to maintain high network availability if a primary link fails. Running multiple active paths simultaneously creates a Layer 2 loop, which causes broadcast storms, MAC address instability, and network performance degradation because Ethernet frames lack a TTL field.

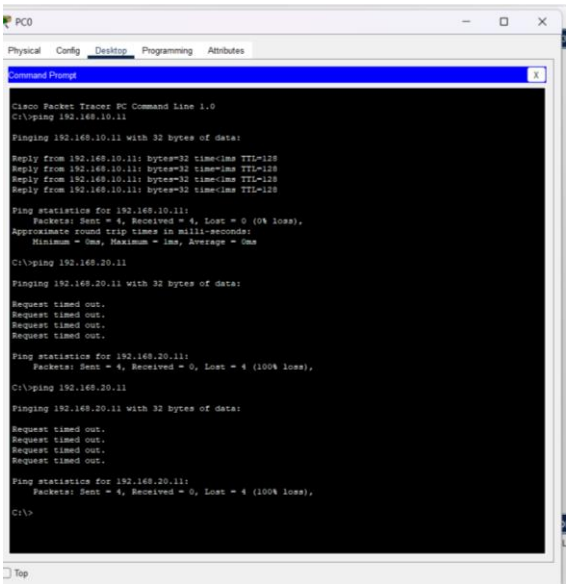
Task 9: Test link failure

Task 10: Troubleshooting


PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.11
Pinging 192.168.10.11 with 32 bytes of data:
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.10.11:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>
Top


PC1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.12
Pinging 192.168.10.12 with 32 bytes of data:
Reply from 192.168.10.12: bytes=32 time=8ms TTL=128
Reply from 192.168.10.12: bytes=32 time=5ms TTL=128
Reply from 192.168.10.12: bytes=32 time=13ms TTL=128
Reply from 192.168.10.12: bytes=32 time=8ms TTL=128
Ping statistics for 192.168.10.12:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 5ms, Maximum = 13ms, Average = 8ms
C:\>
Top





Questions to answer

1. What is Spanning Tree Protocol, and why is it needed in a switched network?

Spanning Tree Protocol (STP) is a Layer 2 network protocol that prevents switching loops in a switched Ethernet network. It creates a loop-free logical topology by blocking redundant links while keeping them available as backups.

2. What is a Layer 2 switching loop? Describe two problems it can cause.

A Layer 2 switching loop happens when data frames keep circulating between switches because there is a loop in the network.

Two problems caused by this are:

1. Broadcast storm: Broadcast messages keep circulating and use a large amount of network bandwidth.

2. MAC address table instability: Switches may learn the same MAC address on different ports, causing frames to be forwarded incorrectly.

3. What is a Root Bridge, and how does STP elect one?

The Root Bridge is the main switch used as a reference point when STP calculates the best paths in a network. STP selects the switch with the lowest Bridge ID as the Root Bridge. If two switches have the same priority, the switch with the lowest MAC address is selected.

4. What is a Bridge ID? Explain the roles of bridge priority and MAC address.

A Bridge ID (BID) is a unique identifier used by STP to select the Root Bridge.

It includes the bridge priority and MAC address. The bridge priority helps determine which switch becomes the Root Bridge. A lower priority is preferred. If the priorities are equal, the switch with the lowest MAC address wins the election.

5. What is the difference between a Root Port and a Designated Port?

Root Port: The port on a non-root switch that provides the lowest-cost path to the Root Bridge.

Designated Port: The port is selected to forward traffic for a particular network segment. Each segment has one Designated Port.

6. What is an Alternate Port, and why might it be placed in a blocking state?

An Alternate Port provides another path to the Root Bridge if the main path fails. It is kept in a blocking state to prevent network loops. If the active path fails, the Alternate Port may start forwarding traffic to restore connectivity.

7. What is STP path cost? How does link speed influence path selection?

STP path cost is a value used to select the best path to the Root Bridge. The cost depends mainly on the speed of the network link. Faster links generally have lower costs, so STP usually prefers a path with a lower total cost.

8. Why must VLANs 10 and 20 be considered separately when using PVST+?

PVST+ runs a separate STP instance for each VLAN. Therefore, VLAN 10 and VLAN 20 can have different Root Bridges and port states.This allows STP to prevent loops separately for each VLAN and helps use redundant network links more efficiently.

9. What happens when an active link fails? How does STP respond?

When an active link fails, STP detects the failure and recalculates the network topology. If another path is available, a previously blocked port may change to a forwarding state. This allows devices to communicate through the backup path.

10. Why is a successful ping not sufficient proof that STP is working correctly?

A successful ping only shows that two devices can communicate at that moment. It does not confirm that STP is working correctly or preventing network loops.Therefore, we should also check the Root Bridge, port roles, port states, and STP information for each VLAN.

11. What is the purpose of BPDUs?

Bridge Protocol Data Units (BPDUs) are messages exchanged between switches to share information about the network. They help STP elect the Root Bridge, select the best paths, determine port roles, and respond to topology changes.

12. What could happen if STP were disabled on all switches while the triangle topology remained connected?

If STP were disabled, all links in the triangle topology could forward traffic and create a switching loop. This could lead to broadcast storms, MAC address table instability, high network traffic, and even a complete network failure.